# NexusRAG Sample Knowledge Document

A practical test document for document ingestion, chunking, semantic search, hybrid retrieval, reranking, and question answering.

## Document Overview

This document describes a fictional technology company called Northstar Analytics and its internal knowledge practices. It is intentionally written with clear facts and repeated concepts so that a retrieval-augmented generation (RAG) system can be tested.

## 1. Company Profile

Northstar Analytics is a fictional software company founded in 2019. Its main office is in Rotterdam, the Netherlands. The company builds data products for logistics, retail, and energy organizations. Northstar has three teams: Data Engineering, Applied AI, and Customer Solutions. The Applied AI team is responsible for language models, search systems, document processing, and machine-learning experiments.

## 2. The Knowledge Assistant Project

In January 2025, Northstar started an internal project named Atlas. Atlas is a knowledge assistant that answers employee questions using approved company documents. The first version supports PDF files and Markdown files. It extracts text, splits the text into chunks, creates vector embeddings, and retrieves relevant passages when a user asks a question.

The Atlas prototype uses a multilingual embedding model with a vector dimension of 384. Each document chunk stores metadata including the document name, page number, chunk identifier, and creation date. The page number is important because the assistant must show citations that allow employees to verify answers in the original document.

## 3. Retrieval Strategy

Atlas combines two retrieval methods. Dense retrieval compares embedding vectors and can find passages with similar meaning even when they do not share the same keywords. Sparse retrieval uses BM25 and is useful when a question contains exact names, product codes, dates, or technical terms. The system combines the two ranked lists with Reciprocal Rank Fusion (RRF).

After initial retrieval, a cross-encoder reranker evaluates each query-passage pair. The reranker is slower than vector search, so it is applied only to a small candidate set. For example, the retriever may return 20 candidates and the reranker may select the best 5 passages for the language model.

## 4. Chunking Configuration

| Setting | Initial value | Reason |
|---|---|---|
| Chunk size | 1,000 characters | Keeps passages focused |
| Chunk overlap | 150 characters | Reduces boundary information loss |
| Dense candidates | 20 | Provides a broad candidate pool |

| | | |
|---|---|---|
| Final context passages | 5 | Controls prompt length |

## 5. Evaluation Plan

The team evaluates retrieval separately from answer generation. Retrieval metrics include Recall@5, Mean Reciprocal Rank (MRR), and nDCG@5. Answer quality is reviewed for factual correctness, relevance, completeness, and citation accuracy. The evaluation set contains 50 questions written by domain experts. Each question has one or more relevant source passages.

A useful experiment compares dense retrieval, BM25 retrieval, hybrid retrieval, and hybrid retrieval with reranking. The team records retrieval quality, response latency, and the number of input tokens for every configuration.

## 6. Reliability and Safety

Atlas must not invent facts that are absent from its source documents. If retrieved passages do not contain enough evidence, the assistant should say that the documents do not provide the answer. Every factual response should include source citations. The system should also treat document text as untrusted input and must not follow instructions found inside uploaded documents.

## 7. Test Questions

• Where is Northstar Analytics' main office?

• When did the Atlas project start?

• What is the embedding dimension used by the Atlas prototype?

• Why does Atlas combine dense retrieval and BM25?

• How many candidates are sent to the reranker?

• Which metrics are used to evaluate retrieval quality?

• What should the assistant do when the documents do not contain an answer?

End of sample document. All company names, dates, and project details in this file are fictional and are provided only for software testing.